SSC MTS 2024 question paper with answers

30 October 2024 · Shift 3

78 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 12 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 13 questions

Q.1

If the average of nine numbers is 360, then what is the sum of the nine numbers?

(A) 3240
(B) 1620
(C) 3420
(D) 40

Q.2

A money lender lent out ₹40,000 in two parts, one at 7% p.a. and the other at 6% p.a.. If the total annual simple interest on the amount is ₹2,550, what is the amount of money lent at 7%?

(A) ₹21,000
(B) ₹26,000
(C) ₹15,000
(D) ₹18,000

Q.3

A bag contains coins of 1 rupee, 0.5 rupee and 0.25 rupee in the ratio of 5: 6: 8. If the total amount of money in the bag is ₹420, find the number of coins of type1 rupee, 0.5 rupee and 0.25 rupee are:

(A) 212,232,312
(B) 210,252,336
(C) 216,252,292
(D) 208,272,352

Q.4

If the price of a commodity is increased by 10% and 20% successively, what is the net percentage increase/decrease in its price, if it is further decreased by 25%?

(A) 5% Increase
(B) 1% Increase
(C) 1% Decrease
(D) 5% Decrease

Q.7

By selling an article for ₹19.5, a dealer makes a profit of 25%. By how much should he increase his selling price (in ₹) so as to make a profit of 50%?

(A) 6.9
(B) 5.9
(C) 3.9
(D) 4.9

Q.8

Rahul sells an article at 8% above its cost price. If he had bought it at 5% less than what he had paid for it and sold it at ₹8 less, he would have gained 12%. Find the cost price of the article.

(A) ₹550
(B) ₹650
(C) ₹700
(D) ₹500

Q.11

The average monthly pocket money of Ram and Shyam is ₹2,460, Shyam and Manohar is ₹2,380 and Manohar and Ram is ₹2,760. Find the monthly pocket money of Ram alone.

(A) ₹2,840
(B) ₹2,048
(C) ₹2,408
(D) ₹2,480

Q.12

A thief is noticed by a policeman from a distance of 400 m. The thief starts running at a speed of 8 km/h and the policeman chases him at a speed of 9 km/h. What is the distance between them after 18 minutes?

(A) 100 m
(B) 150 m
(C) 200 m
(D) 300 m

Q.13

Find the simple interest (in ₹) on ₹2000 as a sum borrowed at 8% per year rate of interest for 6 years.

(A) 960
(B) 900
(C) 1000
(D) 920

Q.15

In Class I there are 12 students of average age 20 years, and in Class II there are 16 students of average age 23 years. What is the average age (approximately) of both the classes?

(A) 17.6 years
(B) 20.7 years
(C) 21.7 years
(D) 15.3 years

Q.17

Two sections A and B of class VIII consist of 42 and 38 students, respectively. If the average marks of section A is 400 and that of section B is 320, find the average marks of the whole class.

(A) 360
(B) 310
(C) 362
(D) 576

Q.18

What sum (in ₹) will earn a simple interest of ₹240 in 2 years at 4% per year rate of interest?

(A) 3000
(B) 1500
(C) 6000
(D) 4000

Q.19

Three bells toll at intervals of 12, 15 and 18 minutes, respectively. If all the three bells toll together at 8 a.m., when will they toll together next?

(A) 8 a.m.
(B) 11 a.m.
(C) 3 p.m.
(D) 10 a.m.

Reasoning Ability and Problem Solving 16 questions

Q.1

What will come in the place of '?' in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 102 – 36 + 59 ÷ 4 × 2 = ?

(A) 15
(B) 20
(C) 5
(D) 10

Q.3

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: All doors are locks. Some doors are keys. Conclusions: (I) Some locks are keys. (II) Some keys are doors.

(A) Neither conclusion I nor II follows
(B) Both conclusions I and II follow
(C) Only conclusion II follows
(D) Only conclusion I follows

Q.4

LJPN is related to SQWU in a certain way based on the English alphabetical order. In the same way, JHNL is related to QOUS. To which of the following is OMSQ related, following the same logic?

(A) TVZX
(B) TVXZ
(C) VTZX
(D) VTXZ

Q.5

In a code language, 'XCMY' is coded as ‘AHPD’ and 'PGVK' is coded as ‘SLYP’. How will 'JOCH' be coded in that language?

(A) PSDN
(B) MTFM
(C) LTDL
(D) NRFK

Q.6

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: All laptops are desktops. No desktop is a computer. Conclusions: (I) No laptop is a computer. (II) Some computers are desktops.

(A) Only conclusion I follows
(B) Neither conclusion I nor II follows
(C) Only conclusion II follows
(D) Both conclusion I and II follow

Q.9

In a certain code language, ‘I like music’ is written as ‘mo ro go’ and ‘can I go’ is writtenas ‘ho mo bo’. How is ‘I’ writtenin the given language?

(A) bo
(B) mo
(C) ho
(D) go

Q.10

In a certain code language, ‘you may laugh’ is coded as ‘CD EJ KO’ and ‘may I go’ is coded as ‘TK PO EJ’. How is ‘may’ coded in that code language?

(A) TK
(B) CD
(C) PO
(D) EJ

Q.11

2 is related to 9 following a certain logic. Following the same logic, 4 is related to 65. To which of the following is 8 related, following the same logic? (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding / subtracting / multiplying, etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 517
(B) 513
(C) 511
(D) 515

Q.12

Seven colleagues (L, M, N, O, P, Q, R) are sitting around a circular meeting table, all facing towards the centre (but not necessarily in the same order). O is sitting to the immediate right of L, while P is sitting to the immediate left of L. Only Q is sitting between R and N, and another one of R's immediate neighbours is M, who is also an immediate neighbour of O. Which two of the following are not immediate neighbours?

(A) R and M
(B) O and R
(C) Q and N
(D) P and N

Q.14

Which two numbers should be interchanged to make the following equation correct? $9 \times 2 - (150 \div 5) + 24 + (32 \div 8) \times 6 + 15 = 71$ (Note: Interchange should be done of the entire number and not individual digits of a given number.)

(A) 9 and 6
(B) 5 and 15
(C) 2 and 6
(D) 5 and 6

Q.15

Select the number from among the given options that can replace the question mark (?) in the following series. 5, 16, 60, 236, 940, ?

(A) 4678
(B) 6899
(C) 678
(D) 3756

Q.16

What should come in place of the question mark (?) in the given series based on the English alphabetical order? TQQ, PRM, LSI, HTE, ?

(A) EUB
(B) DUB
(C) DUA
(D) EUA

Q.17

Lalla, Bablu, Mahesh, Abdul, Arif, Surekha and Kiku are sitting around a circular table, facing the centre (but not necessarily in the same order). Lalla sits 2nd to the right of Bablu. Mahesh sits 3rd to the left of Abdul. Arif sits to the immediate left of Abdul. Surekha is an immediate neighbour of Bablu. Kiku is not an immediate neighbour of Surekha. Who sits third to the right of Abdul?

(A) Lalla
(B) Surekha
(C) Bablu
(D) Kiku

Q.18

POEN is related to LKAJ in a certain way based on the English alphabetical order. In the same way, RADIUS is related to NWZEQO. To which of the following is LOGIC related, following the same logic?

(A) HGCFY
(B) HKCFY
(C) GKCEY
(D) HKCEY

Q.19

11 is related to 57 following a certain logic. Following the same logic, 14 is related to 72. To which of the following is 17 related, following the same logic? (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 89
(B) 85
(C) 87
(D) 83

Q.20

In a certain code language, 'GET' is coded as ‘50’ and 'CAB' is coded as ‘76’. How will 'RUN' be coded in the same language?

(A) 31
(B) 28
(C) 29
(D) 30

General Awareness 25 questions

Q.1

Which of the following country’s official language is Sinhala?

(A) Maldives
(B) Sri Lanka
(C) Nepal
(D) Myanmar

Q.2

Megasthenes was an ambassador who was sent to the court of which Indian ruler by the Greek ruler, Seleucus Nicator?

(A) Chandragupta
(B) Pushyamitra Shunga
(C) Ashoka
(D) Bindusara

Q.3

Which of the following statements is correct?

(A) Chlorine has the highest electron affinity, while fluorine has the highest electronegativity.
(B) Fluorine has the highest electron affinity and the highest electronegativity.
(C) Fluorine has the highest electron affinity, while chlorine has the highest electronegativity.
(D) Fluorine has the highest electron affinity, while bromine has the highest electronegativity.

Q.4

Which country was a major importer of dye intermediates and castor oils from India in 2022-23?

(A) Belgium
(B) Brazil
(C) Japan
(D) China

Q.5

Who has been appointed as the 'UPI Safety Ambassador' by the National Payments Corporation of India (NPCI) to enhance safety awareness on digital payment platforms in November 2023?

(A) Taapsee Pannu
(B) Pankaj Tripathi
(C) Rajkummar Rao
(D) Ayushmann Khurrana

Q.6

In which game are terms like break, cannons, cue, potting, and kissed commonly used?

(A) Polo
(B) Chess
(C) Billiards
(D) Golf

Q.7

Who among the following founded the Servants of India Society in 1905 with the help of MG Ranade?

(A) Jawaharlal Nehru
(B) Mahatma Gandhi
(C) Gopal Krishna Gokhale
(D) Bal Gangadhar Tilak

Q.8

Who is known to have revived Bharatnatyam dance form in its modern version today?

(A) Rukmini Devi
(B) T Balasaraswati
(C) Mallika Sarabhai
(D) Yamini Krishnamurthy

Q.9
Who among the following started the Khilafat Movement against the British government?

(A) Shaukat Ali and Mohammad Ali
(B) Syed Raza Ali and Muhammad Ali Jinnah
(C) Maulana Mazharul Haque and Mian Abdul Aziz
(D) Syed Nabiullah and MA Ansari

Q.10
In which Indian state do we find caves named after elephant statues found in it?

(A) Rajasthan
(B) Karnataka
(C) Odisha
(D) Maharashtra

Q.11
There shall be a/an _____ with the Prime Minister at the head to aid and advise the President in the exercise of his functions.

(A) Attorney General for India
(B) Council of States
(C) Parliament
(D) Council of Ministers

Q.12
When the pH value of soils is more than 7.5, they are considered to be:

(A) neutral soils
(B) alkaline soils
(C) strongly acidic soils
(D) slightly acidic soils

Q.13
Which state in India has the highest population density as per the 2011 census?

(A) Bihar
(B) Tamil Nadu
(C) West Bengal
(D) Uttar Pradesh

Q.14
On 25 August 2023, which State Government expanded its ongoing 'Breakfast Scheme' for students of classes 1 to 5, studying in 31,008 government schools across the State?

(A) Jharkhand
(B) Tamil Nadu
(C) Karnataka
(D) Goa

Q.15
Who among the following authors wrote ‘Kamayani’, ‘Dhruvswamini’, ‘Kankal’, ‘Titli’, ‘Skandagupta’, ‘Lahar’ and ‘Mamta’?

(A) Bhisham Sahni
(B) Jaishankar Prasad
(C) Suryakant Tripathi Nirala
(D) Krishna Sobti

Q.16
Which of the following dance performances begins with the Kelikottu?

(A) Kathakali
(B) Bharatanatyam
(C) Odissi
(D) Kathak

Q.17
The Guindy National Park is located in which state?

(A) Madhya Pradesh
(B) Rajasthan
(C) Tamil Nadu
(D) Maharashtra

Q.18
The budget of the central government is known as the _____.

(A) revised budget
(B) union budget
(C) annual exposed budget
(D) state budget

Q.19
Who among the following Dance Gurus got Padma Vibhushan in the year 2000?

(A) Kelucharan Mohapatra
(B) Bipin Singh
(C) Birju Maharaj
(D) Ileana Citarist

Q.20
The Krebs cycle of aerobic respiration is carried out within which of the following cell organelles?

(A) Lysosome
(B) Mitochondria
(C) Spirogyra
(D) Nucleus

Q.21
Who was sent as an envoy to China during the time of Muhammad bin Tughluq?

(A) Hasan Nizami
(B) Al-Biruni
(C) Ziauddin Barani
(D) Ibn Battuta

Q.22
Which Indian city hosted the Women's Asian Hockey Champions Trophy event in 2023?

(A) New Delhi
(B) Bhopal
(C) Ranchi
(D) Bhubaneshwar

Q.23
What writ means ‘We Command’?

(A) Prohibition
(B) Habeas Corpus
(C) Certiorari
(D) Mandamus

Q.24
The cultural sport of Maharashtra, 'dahi-handi' is associated with which festival?

(A) Ramanavmi
(B) Ganesh Chaturthi
(C) Holi
(D) Krishna Janmashtami

Q.25
Which chemical compound is commonly utilised as a preservative in tomato sauce?

(A) Vinegar
(B) Green Tea Extract
(C) Sodium Benzoate
(D) Carnosic Acid

English Language and Comprehension 24 questions

Q.1
Select the option that will improve the underlined part of the given sentence. The sports teacher saw “across” the game and punished the students who were cheating in the game.

(A) underneath
(B) beside
(C) beneath
(D) through

Q.2
The following sentence has been split into four segments. Identify the segment that contains an error in the usage of indefinite articles. We have invited / our friends / for a / dinner.

(A) our friends
(B) We have invited
(C) dinner
(D) for a

Q.3

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. The company decided / to phase up its / outdated technology / and embrace innovation.

(A) outdated technology
(B) to phase up its
(C) The company decided
(D) and embrace innovation.

Q.4

Select the most appropriate adjective to fill in the blank. After coming from a long journey he said, “It feels so _____ to be home”.

(A) good
(B) best
(C) much good
(D) better

Q.5

Select the INCORRECTLY spelt word..

(A) Gesture
(B) Postuer
(C) Lecture
(D) Culture

Q.6

Select the most appropriate meaning of the underlined idiom in the given sentence. My old car was “pressed into service” as an emergency vehicle for my ailing grandmother.

(A) Asking for special maintenance
(B) Used to take someone to the hospital
(C) Cared for by old people
(D) To use for a particular purpose when a special need occurs.

Q.7

Select the option that can be used as a one-word substitute for the given group of words. A word, number , or other figure that looks the same when viewed (or rotated) upside down .

(A) Anagram
(B) Alliteration
(C) Ambigram
(D) Anadiplosis

Q.8

Select the grammatically correct sentence from the given options.

(A) A hard disc of the computer on the desk needs to be repaired at the earliest.
(B) The hard disc of the computer on the desk needs to be repaired at the earliest.
(C) The hard disc of a computer on a desk needs to be repaired at the earliest.
(D) The hard disc of a computer on the desk needs to be repaired at an earliest.

Q.9

Select the most appropriate ANTONYM of the given word. Piquant

(A) Bitter
(B) Bland
(C) Delectable
(D) Delicious

Q.10

Select the most appropriate synonym of the given word. Fabricate

(A) Fascinate
(B) Construct
(C) Garments
(D) Floral

Q.11

Parts of the following sentence have been given as options. Select the option that contains a grammatical error. If you don't find any error, mark 'No error' as your answer. Mohan is still confused about his higher studies.

(A) Mohan is still
(B) his higher studies
(C) confused about
(D) No error

Q.12

Select the most appropriate option that can substitute the underlined segment in the given sentence. Ramya "would gladly help" Manya if you had asked her.

(A) would have gladly helped
(B) will gladly help
(C) is to gladly help
(D) was to have gladly helped

Q.14

Select the INCORRECTLY spelt word.

(A) Choronology
(B) Protein
(C) Competent
(D) Soliloquy

Q.15

Select the idiom that can substitute the bracketed word segment correctly and complete the given sentence meaningfully. The negotiations reached a point where both parties had to (confront the consequences of their actions) and make tough decisions.

(A) face the music
(B) break a leg
(C) get cold feet
(D) cut corners

Q.16

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. The manager had / thrown up the / drunk people sitting / inside his office premises.

(A) inside his office premises
(B) The manager had
(C) thrown up the
(D) drunk people sitting

Q.17

Select the most appropriate synonym of the given word. Gregarious

(A) Sociable
(B) Introverted
(C) Reserved
(D) Isolated

Q.18

Select the most appropriate synonym of the given word. Fantastic

(A) Delicate
(B) Practical
(C) Fanciful
(D) Graceful

Q.19

Select the most appropriate synonym of the underlined word in the following sentence. You must "adhere" to the rules of this place.

(A) Aware
(B) Strict
(C) Remember
(D) Comply

Q.20

Select the INCORRECTLY spelt word…

(A) Masquerade
(B) Squadroun
(C) Caricature
(D) Vigilante

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Man, it would seem, has (1)_____ from arboreal apes. They lived a happy life in tropical forests, eating coconuts when they were hungry, and throwing them at each other when they were not. They were (2)_____ occupied in gymnastics and acquired an (3)_____ which to us is truly (4)_____. But after some millions of years of this arboreal paradise, their numbers increased to the point where the supply of coconuts was no longer (5)_____.

Select the most appropriate option to fill in blank number 1.

(A) extended
(B) ascended
(C) descended
(D) recommended

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Man, it would seem, has (1)_____ from arboreal apes. They lived a happy life in tropical forests, eating coconuts when they were hungry, and throwing them at each other when they were not. They were (2)_____ occupied in gymnastics and acquired an (3)_____ which to us is truly (4)_____. But after some millions of years of this arboreal paradise, their numbers increased to the point where the supply of coconuts was no longer (5)_____.

Select the most appropriate option to fill in blank number 2.

(A) professionally
(B) perpetually
(C) bisectionally
(D) conspiratorially

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Man, it would seem, has (1)_____ from arboreal apes. They lived a happy life in tropical forests, eating coconuts when they were hungry, and throwing them at each other when they were not. They were (2)_____ occupied in gymnastics and acquired an (3)_____ which to us is truly (4)_____. But after some millions of years of this arboreal paradise, their numbers increased to the point where the supply of coconuts was no longer (5)_____.

Select the most appropriate option to fill in blank number 3.

(A) variability
(B) density
(C) civility
(D) agility

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Man, it would seem, has (1)_____ from arboreal apes. They lived a happy life in tropical forests, eating coconuts when they were hungry, and throwing them at each other when they were not. They were (2)_____ occupied in gymnastics and acquired an (3)_____ which to us is truly (4)_____. But after some millions of years of this arboreal paradise, their numbers increased to the point where the supply of coconuts was no longer (5)_____.

Select the most appropriate option to fill in blank number 4.

(A) disappointing
(B) frightening
(C) astonishing
(D) boring

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Man, it would seem, has (1)_____ from arboreal apes. They lived a happy life in tropical forests, eating coconuts when they were hungry, and throwing them at each other when they were not. They were (2)_____ occupied in gymnastics and acquired an (3)_____ which to us is truly (4)_____. But after some millions of years of this arboreal paradise, their numbers increased to the point where the supply of coconuts was no longer (5)_____.

Select the most appropriate option to fill in blank number 5.

(A) delicate

(B) consecrate

(C) adequate

(D) deficient

Answer key

Numerical and Mathematical Ability

Q.1 **(A)**	Q.2 **(C)**	Q.3 **(B)**	Q.4 **(C)**	Q.7 **(C)**	Q.8 **(D)**
Q.11 **(A)**	Q.12 **(A)**	Q.13 **(A)**	Q.15 **(C)**	Q.17 **(C)**	Q.18 **(A)**
Q.19 **(B)**					

Reasoning Ability and Problem Solving

Q.1 **(B)**	Q.3 **(B)**	Q.4 **(C)**	Q.5 **(B)**	Q.6 **(A)**	Q.9 **(B)**
Q.10 **(D)**	Q.11 **(B)**	Q.12 **(B)**	Q.14 **(C)**	Q.15 **(D)**	Q.16 **(C)**
Q.17 **(A)**	Q.18 **(D)**	Q.19 **(C)**	Q.20 **(C)**		

General Awareness

Q.1 **(B)**	Q.2 **(A)**	Q.3 **(A)**	Q.4 **(D)**	Q.5 **(B)**	Q.6 **(C)**
Q.7 **(C)**	Q.8 **(A)**	Q.9 **(A)**	Q.10 **(D)**	Q.11 **(D)**	Q.12 **(B)**
Q.13 **(A)**	Q.14 **(B)**	Q.15 **(B)**	Q.16 **(A)**	Q.17 **(C)**	Q.18 **(B)**
Q.19 **(A)**	Q.20 **(B)**	Q.21 **(D)**	Q.22 **(C)**	Q.23 **(D)**	Q.24 **(D)**
Q.25 **(C)**					

English Language and Comprehension

Q.1 **(D)**	Q.2 **(D)**	Q.3 **(B)**	Q.4 **(A)**	Q.5 **(B)**	Q.6 **(D)**
Q.7 **(C)**	Q.8 **(B)**	Q.9 **(B)**	Q.10 **(B)**	Q.11 **(D)**	Q.12 **(A)**
Q.14 **(A)**	Q.15 **(A)**	Q.16 **(C)**	Q.17 **(A)**	Q.18 **(C)**	Q.19 **(D)**
Q.20 **(B)**	Q.21 **(C)**	Q.22 **(B)**	Q.23 **(D)**	Q.24 **(C)**	Q.25 **(C)**

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (A) 3240

Sum = average × count = 360 × 9 = 3,240.

Q.2 · Answer (C) ₹15,000

If x is lent at 7%, 0.07x + 0.06(40,000 − x) = 2,550, so 0.01x = 2,550 − 2,400 = 150 and x = ₹15,000.

Q.3 · Answer (B) 210,252,336

Each set of 5, 6 and 8 coins is worth 5 × 1 + 6 × 0.5 + 8 × 0.25 = ₹10, so there are 420/10 = 42 sets: 5 × 42 = 210, 6 × 42 = 252 and 8 × 42 = 336 coins.

Q.4 · Answer (C) 1% Decrease

1.1 × 1.2 × 0.75 = 0.99 of the original price: a net decrease of 1%.

Q.7 · Answer (C) 3.9

Cost price = 19.5/1.25 = ₹15.6. A 50% profit needs 15.6 × 1.5 = ₹23.4, so the price must rise by 23.4 − 19.5 = ₹3.9.

Q.8 · Answer (D) ₹500

With cost price x: 1.08x − 8 = 1.12 × 0.95x = 1.064x, so 0.016x = 8 and x = ₹500.

Q.11 · Answer (A) ₹2,840

The three pairs total ₹4,920, ₹4,760 and ₹5,520, which together count each person twice: 15,200/2 = ₹7,600 for all three. Ram = 7,600 − 4,760 (Shyam and Manohar) = ₹2,840.

Q.12 · Answer (A) 100 m

The policeman gains 9 − 8 = 1 km/h, so in 18 minutes (0.3 hour) he closes 0.3 km = 300 m of the 400 m gap, leaving 100 m.

Q.13 · Answer (A) 960

SI = 2,000 × 8 × 6/100 = ₹960.

Q.15 · Answer (C) 21.7 years

(12 × 20 + 16 × 23)/28 = (240 + 368)/28 = 608/28 ≈ 21.7 years.

Q.17 · Answer (C) 362

(42 × 400 + 38 × 320)/80 = (16,800 + 12,160)/80 = 28,960/80 = 362.

Q.18 · Answer (A) 3000

P = SI × 100/(R × T) = 240 × 100/(4 × 2) = ₹3,000.

Q.19 · Answer (B) 11 a.m.

The bells toll together every LCM(12, 15, 18) = 180 minutes, that is 3 hours, so next at 11 a.m.

Reasoning Ability and Problem Solving

Q.1 · Answer (B) 20

After interchanging the signs: 102 + 36 − 59 × 4 ÷ 2 = 102 + 36 − 118 = 20.

Q.3 · Answer (B) Both conclusions I and II follow

The doors that are keys are also locks, so some locks are keys (I); 'some doors are keys' converts to 'some keys are doors' (II). Both follow.

Q.4 · Answer (C) VTZX

Each letter moves 7 places forward: LJPN → SQWU and JHNL → QOUS. So OMSQ → VTZX.

Q.5 · Answer (B) MTFM

The letters move +3, +5, +3, +5 (wrapping round after Z): XCMY → AHPD and PGVK → SLYP. So JOCH → MTFM.

Q.6 · Answer (A) Only conclusion I follows
Every laptop is a desktop and no desktop is a computer, so no laptop is a computer (I). II contradicts 'no desktop is a computer', so it does not follow.

Q.9 · Answer (B) mo
'I' is the only word in both sentences and 'mo' the only code in both, so I = mo.

Q.10 · Answer (D) EJ
'may' is the only word in both sentences and 'EJ' the only code in both, so may = EJ.

Q.11 · Answer (B) 513
Each number is related to its cube plus 1: $2^3 + 1 = 9$ and $4^3 + 1 = 65$. Likewise $8^3 + 1 = 513$.

Q.12 · Answer (B) O and R
L sits between P and O, Q between R and N, and M between R and O. Round the table the order is P, L, O, M, R, Q, N, and back to P. O's neighbours are L and M, so O and R are not neighbours.

Q.14 · Answer (C) 2 and 6
Interchanging 2 and 6: $9 \times 6 - (150 \div 5) + 24 + (32 \div 8) \times 2 + 15 = 54 - 30 + 24 + 8 + 15 = 71$.

Q.15 · Answer (D) 3756
Each term is 4 times the one before minus 4: $5 \times 4 - 4 = 16$, $16 \times 4 - 4 = 60$ and so on, so the next is $940 \times 4 - 4 = 3756$.

Q.16 · Answer (C) DUA
The first letters move back 4 (T, P, L, H, so D), the second forward 1 (Q, R, S, T, so U) and the third back 4 (Q, M, I, E, so A): DUA.

Q.17 · Answer (A) Lalla
Going round to the left from Abdul the order is Abdul, Arif, Kiku, Mahesh, Lalla, Surekha, Bablu. Arif and Mahesh are fixed by Abdul; Lalla second to Bablu's right puts Bablu either to Abdul's immediate right or third to his right, and the second would leave Surekha and Kiku side by side. Third to Abdul's right is Lalla.

Q.18 · Answer (D) HKCEY
Each letter moves 4 places back (wrapping round before A): POEN → LKAJ and RADIUS → NWZEQO. So LOGIC → HKCEY.

Q.19 · Answer (C) 87
The second number is 5 times the first plus 2: $11 \times 5 + 2 = 57$ and $14 \times 5 + 2 = 72$. Likewise $17 \times 5 + 2 = 87$.

Q.20 · Answer (C) 29
The code is the sum of the letters' places counted from Z (Z = 1, A = 26) plus 1: GET = 20 + 22 + 7 + 1 = 50 and CAB = 24 + 26 + 25 + 1 = 76. So RUN = 9 + 6 + 13 + 1 = 29.

General Awareness

Q.1 · Answer (B) Sri Lanka
Sinhala, with Tamil, is an official language of Sri Lanka.

Q.2 · Answer (A) Chandragupta
Seleucus Nicator sent Megasthenes as his ambassador to Chandragupta Maurya's court at Pataliputra; his 'Indica' describes the Mauryan state.

Q.3 · Answer (A) Chlorine has the highest electron affinity, while fluorine has the highest electronegativity.
Fluorine is the most electronegative element, but chlorine has the higher electron affinity: fluorine's very small atom crowds its electrons, which repel an incoming one.

Q.4 · Answer (D) China
SSC's key gives China, which buys both from India: it is the largest market for India's castor oil (India supplies most of the world's) and a leading one for Indian dyes and dye intermediates.

Q.5 · Answer (B) Pankaj Tripathi
In November 2023 NPCI named the actor Pankaj Tripathi its 'UPI Safety Ambassador', to spread awareness of safe digital payments.

Q.6 · Answer (C) Billiards
Break, cannon, cue, pot and kiss are billiards terms: a cannon is a shot in which the cue ball strikes both the other balls, and a kiss is a touch between two balls.

Q.7 · Answer (C) Gopal Krishna Gokhale
Gopal Krishna Gokhale, whose mentor was M G Ranade, founded the Servants of India Society at Pune in 1905 to train workers for national service.

Q.8 · Answer (A) Rukmini Devi
Rukmini Devi Arundale, who founded Kalakshetra in Chennai in 1936, gave Bharatanatyam the stage form it has today and won it wide respectability.

Q.9 · Answer (A) Shaukat Ali and Mohammad Ali
The Ali brothers, Shaukat Ali and Mohammad Ali, led the Khilafat Movement (1919-24) in defence of the Ottoman Caliph.

Q.10 · Answer (D) Maharashtra
The Elephanta Caves near Mumbai, Maharashtra, were named by the Portuguese after a large stone elephant they found on the island.

Q.11 · Answer (D) Council of Ministers
Article 74(1) provides the Council of Ministers, headed by the Prime Minister, to aid and advise the President, who acts on its advice.

Q.12 · Answer (B) alkaline soils
Soils with a pH below about 6.5 are acidic, those from 6.5 to 7.5 neutral and those above 7.5 alkaline.

Q.13 · Answer (A) Bihar
Bihar had the highest population density in the 2011 census, 1,106 persons per km², ahead of West Bengal.

Q.14 · Answer (B) Tamil Nadu
Tamil Nadu's Chief Minister's Breakfast Scheme, begun in 2022, was extended on 25 August 2023 to all its government primary schools.

Q.15 · Answer (B) Jaishankar Prasad
Jaishankar Prasad, a leading poet of the Chhayavad school of Hindi poetry, wrote the epic 'Kamayani', the plays 'Dhruvswamini' and 'Skandagupta', and the novels 'Kankal' and 'Titli'.

Q.16 · Answer (A) Kathakali
A Kathakali performance opens with the Kelikottu, drumming in the evening to announce that a play will be staged that night.

Q.17 · Answer (C) Tamil Nadu
Guindy National Park lies within the city of Chennai, Tamil Nadu.

Q.18 · Answer (B) union budget
The central government's budget is the Union Budget, which the Constitution (Article 112) calls the Annual Financial Statement.

Q.19 · Answer (A) Kelucharan Mohapatra
Kelucharan Mohapatra, the Odissi guru, received the Padma Vibhushan in 2000; Birju Maharaj had received it in 1986.

Q.20 · Answer (B) Mitochondria
The Krebs (citric acid) cycle runs in the matrix of the mitochondria. Spirogyra is an alga, not an organelle.

Q.21 · Answer (D) Ibn Battuta
The Moroccan traveller Ibn Battuta served Muhammad bin Tughluq as qazi of Delhi and was sent by him as envoy to the Chinese emperor in 1342.

Q.22 · Answer (C) Ranchi
Ranchi, Jharkhand, hosted the 2023 Women's Asian Champions Trophy, which India won.

Q.23 · Answer (D) Mandamus
'Mandamus' is Latin for 'we command': the court orders a public official or body to perform a legal duty.

Q.24 · Answer (D) Krishna Janmashtami
Dahi Handi, in which a human pyramid breaks a pot of curd hung high, recalls the boy Krishna's love of butter and is part of the Krishna Janmashtami (Gokulashtami) celebrations.

Q.25 · Answer (C) Sodium Benzoate
Sodium benzoate is a common chemical preservative in sauces, jams and squashes. Vinegar also preserves food, but it is a solution of acetic acid, not a single compound.

English Language and Comprehension

Q.1 · Answer (D) through
To 'see through' something is to recognise the deceit behind it: the teacher 'saw through' the game and punished the cheats.

Q.2 · Answer (D) for a
Meals usually take no article: 'We have invited our friends for dinner.'

Q.3 · Answer (B) to phase up its
To stop using something gradually is to 'phase it out': 'to phase out its outdated technology'.

Q.4 · Answer (A) good
'So' takes the positive degree: 'It feels so good to be home.'

Q.5 · Answer (B) Postuer
The correct spelling is 'posture'.

Q.6 · Answer (D) To use for a particular purpose when a special need occurs.
To 'press something into service' is to put it to use for a purpose when a special need arises.

Q.7 · Answer (C) Ambigram
An ambigram reads the same when turned upside down. An anagram rearranges a word's letters, and alliteration repeats the opening sounds of words.

Q.8 · Answer (B) The hard disc of the computer on the desk needs to be repaired at the earliest.
SSC's key gives the sentence with 'the' throughout: a repair concerns one particular computer, on a known desk, and its hard disc. 'A hard disc of the computer' (as if it had several) and 'a computer on a desk' are not ungrammatical, but they leave unclear which is meant; 'at an earliest' is wrong, as the fixed phrase is 'at the earliest'.

Q.9 · Answer (B) Bland
'Piquant' means pleasantly sharp or spicy; its opposite is 'bland'.

Q.10 · Answer (B) Construct
To 'fabricate' is to make or 'construct' something (it can also mean to invent something false).

Q.11 · Answer (D) No error
'Is still confused about' is correct in tense and preposition, so the sentence has no error.

Q.12 · Answer (A) would have gladly helped
An 'if' clause in the past perfect ('had asked') takes 'would have' + past participle: 'Ramya would have gladly helped Manya if you had asked her.'

Q.14 · Answer (A) Choronology
The correct spelling is 'chronology'.

Q.15 · Answer (A) face the music
To 'face the music' is to accept the unpleasant consequences of one's actions. To 'get cold feet' is to lose one's nerve, and to 'cut corners' to do something cheaply or carelessly.

Q.16 · Answer (C) thrown up the
To force people to leave is to 'throw them out': 'had thrown out the drunk people'. 'Throw up' means to vomit or to produce.

Q.17 · Answer (A) Sociable
'Gregarious' means fond of company: 'sociable'. The other three are nearer its opposite.

Q.18 · Answer (C) Fanciful
'Fantastic' in its first sense means imaginative and remote from reality: 'fanciful'.

Q.19 · Answer (D) Comply
To 'adhere to' rules is to follow them: to 'comply' with them.

Q.20 · Answer (B) Squadroun
The correct spelling is 'squadron'.

Q.21 · Answer (C) descended
Man has 'descended' from arboreal apes, that is, he is their descendant.

Q.22 · Answer (B) perpetually
They were 'perpetually', that is constantly, occupied in gymnastics.

Q.23 · Answer (D) agility
Constant gymnastics gave them 'agility', nimbleness of movement.

Q.24 · Answer (C) astonishing
An agility which to us is truly 'astonishing', that is amazing.

Q.25 · Answer (C) adequate
As their numbers grew, the supply of coconuts was no longer 'adequate', that is enough.

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